Lightweight Prompt Injection Detection: From Centroid-Based Embedding Classification to a Hybrid Guardrail

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Abstract

Prompt injection attacks against large language models demand robust real-time defenses. This paper investigates three questions: (1) whether embedding representations contain discriminative information for prompt injection detection, (2) why nearest-centroid classification fails on these embeddings, and (3) whether lightweight statistical features offer a viable alternative. Through controlled experiments across embedding models on 144,569 test samples, we find that the embedding space contains discriminative information (linear probes achieve AUC 0.844–0.855) that nearest-centroid scoring cannot access (AUC 0.547). The critical factor is the number of centroids: 4 category centroids overlap ($C = 0.986$) while 2 class centroids achieve AUC 0.763–0.784. We then evaluate a lightweight hybrid guardrail combining 117 handcrafted features, TF-IDF similarity, and a stacking ensemble. The hybrid model achieves 83.0% in-distribution accuracy with a measured single-threaded CPU inference latency of 0.149 ms per prompt. An ablation study reveals that TF-IDF alone (85.0% accuracy) outperforms the full hybrid, indicating that, under the evaluated setting, the 117 handcrafted features do not provide complementary predictive value beyond TF-IDF. Cross-dataset evaluation on four external benchmarks reveals a severe generalization gap: performance drops from 83.0% to 51.5% (a 31.5 percentage-point decline). We present a detailed analysis of the model's strengths, weaknesses, and failure modes, providing experimentally grounded guidance for designing prompt injection detection systems.

Keywords: prompt injection, LLM security, nearest-centroid, linear probe, hybrid guardrail, lightweight classifier, cross-dataset generalization

1 Introduction

The deployment of large language models (LLMs) in production systems has created an urgent security challenge: prompt injection attacks (Schulhoff et al., 2023; Greshake et al., 2023). These adversarial inputs attempt to override system instructions, extract confidential prompts, or bypass safety alignments (Liu et al., 2023). Unlike traditional software vulnerabilities, prompt injections exploit the fundamental architecture of autoregressive models, where user input and system instructions share the same processing context.

An intuitive defense is to use embedding models to compare inputs against known attack corpora via semantic similarity. This paper investigates whether this approach can work, why it fails under certain configurations, and whether lightweight statistical features offer a viable alternative. We address three research questions:

1. **RQ1: Does the embedding representation contain discriminative information for prompt injection detection?**

   Under the evaluated dataset, yes. Linear probes trained on the same embeddings used for centroid scoring achieve AUC 0.844–0.884, indicating that discriminative structure exists in the embedding space even when centroid-based methods fail to exploit it.

2. **RQ2: Why does the original centroid-based classifier fail?**

   The results indicate that the failure is primarily associated with the classification method rather than an absence of discriminative information in the evaluated embeddings. Four category centroids overlap in cosine similarity space ($C = 0.986$), preventing effective nearest-centroid classification. Reducing to two class centroids (malicious vs. benign) improves AUC from 0.547 to 0.763–0.784, and learned linear boundaries perform better still (AUC 0.844–0.884).

3. **RQ3: Can lightweight statistical features provide a computationally efficient alternative?**

   A hybrid guardrail combining 117 handcrafted features, TF-IDF similarity, and a stacking ensemble achieves 83.0% in-distribution accuracy with a measured single-threaded CPU inference latency of 0.149 ms per prompt. However, an ablation study reveals that TF-IDF alone achieves 85.0% accuracy, exceeding the full hybrid, suggesting that handcrafted security features do not complement distributional representations under the evaluated setting. Cross-dataset evaluation on four external benchmarks reveals a severe generalization gap: performance drops to 51.5% (a 31.5 percentage-point decline), suggesting that strong in-distribution results do not necessarily indicate robustness.

Our contribution is threefold:

1. We systematically investigate why nearest-centroid scoring fails on the evaluated embedding representations, showing that the 4-centroid max-similarity configuration performs poorly despite the presence of discriminative information accessible to learned classifiers.

2. We demonstrate that linear probes and 2-centroid classifiers recover discriminative structure from the same embeddings, indicating the embedding space contains security-relevant information.

3. We present a lightweight hybrid guardrail with detailed strength/weakness analysis, including an ablation study showing TF-IDF alone outperforms the full hybrid, and a severe cross-dataset generalization gap (31.5 percentage-point decline).

2 Related Work

Prompt injection attacks exploit the shared context of user input and system instructions in autoregressive language models (Schulhoff et al., 2023; Greshake et al., 2023; Liu et al., 2023). Attack types include direct injection, indirect injection, system prompt extraction, jailbreak (Perez and Ribeiro, 2023; Wei et al., 2023), and refusal bypass (Li et al., 2024).

Defense approaches include instruction hierarchy (Wallace et al., 2024), perplexity-based detection (Alon and Kamfonas, 2023), and retrieval-augmented systems using embedding similarity (Reimers and Gurevych, 2019; Xiao et al., 2023). Recent work has explored LLM-based classifiers and feature-engineered models for injection detection, though most evaluations remain in-distribution (Liu et al., 2023). In representation learning, Khosla et al. (2020) demonstrated supervised contrastive learning, and Jing et al. (2022) identified dimensional collapse. Our work examines a related phenomenon in the security domain: substantial overlap between class centroids in embedding space ($C = 0.986$), which is associated with poor nearest-centroid classification performance, even though trained linear probes can extract discriminative structure.

3 Dataset

All experiments use the Guardrailer Dataset v1, a curated corpus of 722,842 labeled prompts publicly available at https://www.kaggle.com/datasets/prashannadeveloper/guardrailer-dataset

Table 1: Dataset composition by attack category.

Category	**Count**	**% of Total**
benign_control	289,776	40.1%
jailbreak	267,091	36.9%
direct_injection	92,206	12.8%
system_prompt_extraction	73,769	10.2%
Total	**722,842**	**100%**

The dataset is moderately imbalanced: 59.9% malicious (433,066 samples) and 40.1% benign (289,776 samples). Data quality procedures included deduplication (removed 210,138 duplicates), label reconciliation (fixed 114,400 rows), dead column removal, length normalization, extreme length filtering (<5 or >20K chars), and jailbreak category downsampling (20%).

4 The Embedding Failure

The embedding pipeline was designed as a routine validation step: test the embedding approach on the corrected 722K dataset with proper train-test separation. Instead, it revealed significant performance limitations.

The embedding scoring pipeline used bge-large-en-v1.5 (Xiao et al., 2023) to convert prompts into 1024-dimensional embeddings, then computed 7 weighted signals: $s_{\text{composite}} = 0.30 \cdot s_{\text{sparse_idf}} + 0.25 \cdot s_{\text{centroid}} + 0.10 \cdot s_{\text{perplexity}} + 0.10 \cdot s_{\text{entropy}} + 0.10 \cdot s_{\text{token_freq}} + 0.10 \cdot s_{\text{ngram}} + 0.05 \cdot s_{\text{length}}$. The centroid signal computes maximum cosine similarity (Duda et al., 2001) across 4 category centroids: $s_{\text{centroid}} = \max_{c \in \mathcal{C}} \cos(\mathbf{x}, \mu_c)$. Classification is threshold-based: $y = \mathbb{1}[s_{\text{composite}} \geq \tau]$.

4.1 Experimental Setup

- **Dataset:** 722,842 samples, 70/10/20 stratified split (514,020 train / 64,253 val / 144,569 test)

- **Embedding model:** BAAI/bge-large-en-v1.5 (335M parameters, 1024-dim)
- **Classification:** Threshold-based composite score
- **Metrics:** Balanced Accuracy, F1, MCC, AUC-ROC, confusion matrix

4.2 Results

Table 2: Embedding nearest-centroid composite score on 144,569 test samples.

Metric	Value
Balanced Accuracy	0.5005
F1 Score	0.0088
MCC	0.0085
Precision	0.6649
Recall	0.0044
FPR	0.33%
FNR	99.56%
AUC-ROC	0.5466 (0.547 rounded)

The confusion matrix reveals the core failure:

$$\mathbf{C} = \begin{pmatrix} 57{,}763 & 193 \\ 86{,}230 & 383 \end{pmatrix} = \begin{pmatrix} \text{TN} & \text{FP} \\ \text{FN} & \text{TP} \end{pmatrix} \tag{1}$$

At threshold 0.50, the system correctly identifies 99.67% of benign prompts but misses 99.56% of malicious prompts—effectively a "classify everything as safe" system.

4.3 Signal Discrimination Analysis

All seven embedding-derived signals showed weak class separation; the strongest, length normalization, achieved only $\delta = 0.036$—less than half the separation needed for moderate discrimination.

4.4 Threshold Analysis

No threshold provides an acceptable trade-off; AUC = 0.547 and the default threshold yields 99.56% FNR.

5 Investigation: Why the Embedding Method Failed

The embedding failure raised a critical question: does the embedding space truly lack discriminative information, or is the failure caused by the classification method? We conducted four experiments to answer this. All investigation experiments use a 10,000-sample stratified subset (8,000 train / 2,000 test) to enable rapid iteration across four embedding models; the hybrid guardrail experiments use the full dataset.

5.1 Experiment 1: Two-Centroid Cosine-Similarity Classification

We tested whether reducing to 2 centroids (malicious vs. benign) improves performance:

$$\hat{y} = \mathbb{1}[\cos(\mathbf{x}, \mu_{\text{mal}}) > \cos(\mathbf{x}, \mu_{\text{ben}})] \tag{2}$$

On 10,000 stratified samples (8,000 train / 2,000 test), this achieved AUC-ROC 0.763 (Table 3).

Table 3: Two-centroid cosine-similarity classification (10,000 samples).

Model	Params	Centroid C	Accuracy	F1	AUC-ROC
bge-large	335M	0.986	71.8%	0.720	0.772
MiniLM-L6	22M	0.878	69.5%	0.697	0.753
e5-small	33M	0.996	70.8%	0.710	0.769
bge-small	33M	0.987	70.0%	0.703	0.757
Mean	—	—	**70.5%**	**0.708**	**0.763**

The improvement from 0.5 to 0.76 suggests that binary classification with 2 centroids captures discriminative information that 4-centroid max-similarity cannot access.

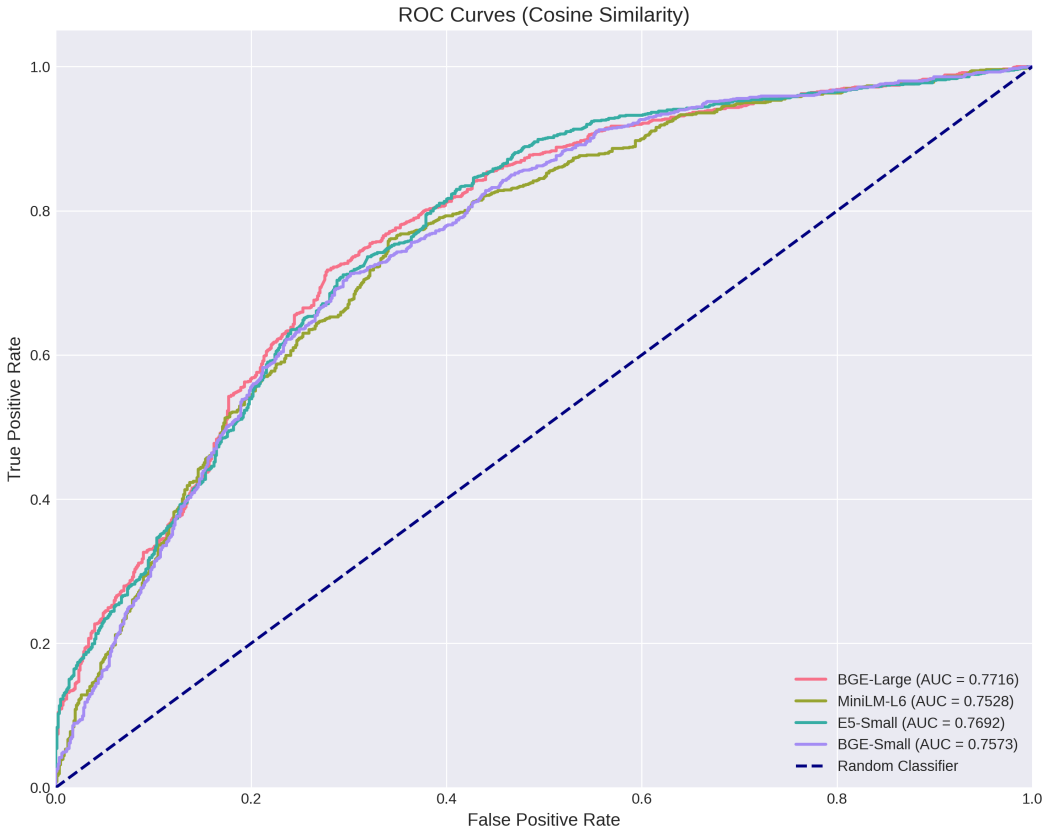


Figure 1: ROC curves for four embedding models using two-centroid cosine-similarity classification. BGE-Large achieves the highest AUC (0.772), followed by E5-Small (0.769), BGE-Small (0.757), and MiniLM-L6 (0.753). All models substantially outperform the random classifier (dashed line).

5.2 Experiment 2: Euclidean Distance with StandardScaler

We tested whether StandardScaler (Duda et al., 2001) + Euclidean distance outperforms cosine similarity:

$$d^{\text{scaled}}(\mathbf{x}, \mathbf{c}) = \sqrt{\sum_{j=1}^{d} \frac{(x_j - c_j)^2}{\sigma_j^2}} \tag{3}$$

where σ_j^2 is the variance of dimension j across all samples. This creates a weighted Euclidean distance where low-variance dimensions get higher weight ($w_j = 1/\sigma_j^2$).

Table 4: Euclidean distance + StandardScaler classification (10,000 samples).

Model	Params	Centroid $\mu \cdot \mu$	Accuracy	F1	AUC-ROC
bge-large	335M	0.468	71.2%	0.714	0.784
MiniLM-L6	22M	0.089	70.0%	0.702	0.755
e5-small	33M	0.791	71.6%	0.718	0.775
bge-small	33M	0.510	70.8%	0.710	0.770
Mean	—	—	**70.9%**	**0.711**	**0.771**

StandardScaler + Euclidean achieves mean AUC 0.771, slightly better than cosine similarity (0.763).

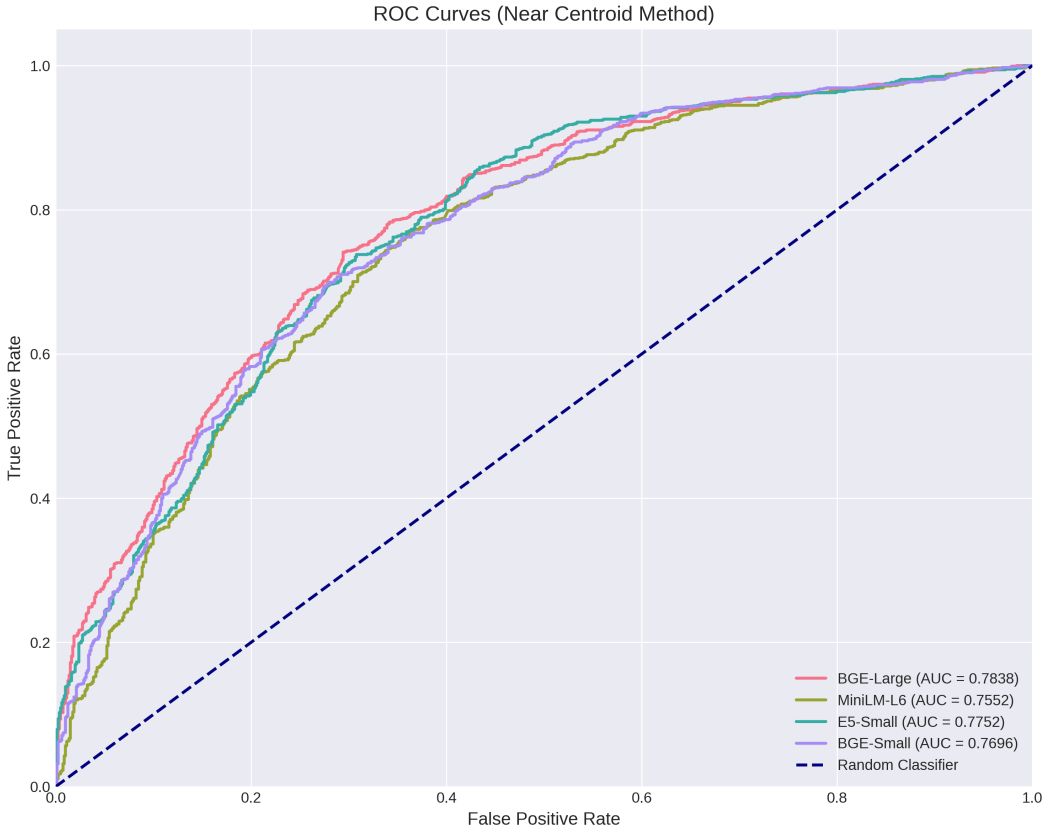


Figure 2: ROC curves for four embedding models using Euclidean distance with StandardScaler. BGE-Large achieves the highest AUC (0.784), followed by E5-Small (0.775), BGE-Small (0.770), and MiniLM-L6 (0.755).

5.3 Experiment 3: Linear Probe (Logistic Regression)

We trained a logistic regression classifier (Duda et al., 2001) on the raw embeddings:

$$P(y = 1|\mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b) \tag{4}$$

Table 5: Linear probe performance (10,000 samples, 2,000 test).

Model	Params	Centroid $\mu \cdot \mu$	Accuracy	F1	AUC-ROC
bge-large	335M	0.470	80.8%	0.845	0.884
e5-small	33M	0.791	80.0%	0.838	0.860
bge-small	33M	0.515	78.0%	0.824	0.855
MiniLM-L6	22M	0.091	77.1%	0.814	0.844

All three models achieve AUC-ROC > 0.84 with linear probes, suggesting that the embedding space contains strong discriminative information.

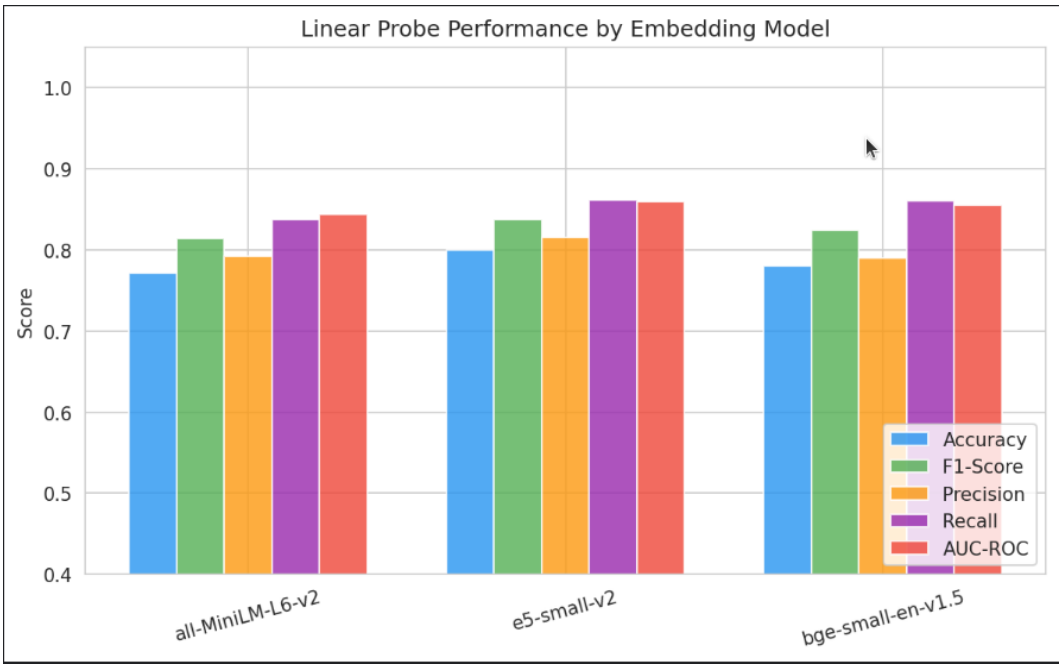


Figure 3: Linear probe performance by embedding model. All three small models achieve AUC-ROC above 0.84, with e5-small-v2 showing the highest AUC and MiniLM-L6 the lowest. The consistent performance across models suggests that discriminative information is present in the evaluated embedding spaces.

5.4 Experiment 4: Comparison of All Methods

Table 6: Comparison of classification methods on bge-large-en-v1.5 embeddings. Row 1 uses the full 722K dataset (70/10/20 split); Rows 2–5 use a 10,000-sample stratified subset (80/20 split). All subsets preserve the original class distribution via stratified sampling.

Method	**Acc.**	**AUC**	**Centroid C**	**Notes**
4-centroid nearest-centroid†	40.2%	0.547	0.986	FAILS
4-centroid nearest-centroid	68.3%	0.558	0.909	FAILS
2-centroid cosine similarity	71.8%	0.772	0.986	Works
Euclidean + StandardScaler	71.2%	0.784	0.468	Works
Linear probe (LogReg)	**80.8%**	**0.884**	—	Best
ML on 7 signals (RF)‡	78.5%	0.873	—	Works

† Full 722K dataset (144,569 test samples). Row 2: 10,000 stratified samples (2,000 test).
‡ Balanced Accuracy. Random Forest trained on 7 embedding weighted signals (sparse_idf, centroid, perplexity, entropy, token_freq, ngram, length) with 5-fold CV.

The linear probe substantially outperforms all centroid-based methods, while the two-centroid approaches achieve intermediate performance. The 4-centroid configuration achieves 40.2% on the full 722K dataset and 68.3% on the 10K subset—both below practical thresholds—confirming that category-level centroid overlap is catastrophic regardless of sample size. The 10K result overestimates true performance: centroids computed from fewer samples are noisier estimates, artificially reducing the measured inter-centroid similarity ($C = 0.909$ vs. $C = 0.986$ on 722K) and shifting decision boundaries to appear more favorable. The 722K result is the more reliable estimate of nearest-centroid performance. The linear probe (LogisticRegression, $C = 1.0$, max_iter=1000) and the Random Forest baseline operate on fundamentally different inputs: the probe learns a decision boundary directly on 1024-dimensional embeddings, while the RF operates on 7 pre-computed weighted similarity signals derived from those same embeddings.

5.5 The Critical Discovery: Why 4 Centroids Fail While 2 Centroids Work

The embedding method assigns each test prompt to the nearest category centroid using cosine similarity:

$$\hat{y} = \arg\max_{c \in \mathcal{C}} \cos(\mathbf{x}, \mu_c) = \arg\max_{c \in \mathcal{C}} \frac{\mathbf{x} \cdot \mu_c}{\|\mathbf{x}\| \|\mu_c\|} \tag{5}$$

where $\mathcal{C} = \{\text{benign}, \text{jailbreak}, \text{injection}, \text{extraction}\}$ and $\mu_c = \frac{1}{|S_c|} \sum_{i \in S_c} \mathbf{x}_i$ is the mean embedding (centroid) of category c computed from the training set.

With 4 centroids, the classifier produces a 4-class prediction which is collapsed to binary: $\hat{y}_{\text{binary}} = \mathbb{1}[\hat{y} \neq \text{benign}]$. The inter-centroid cosine similarity is:

$$C = \frac{1}{|\mathcal{C}|(|\mathcal{C}| - 1)} \sum_{i \neq j} \cos(\mu_i, \mu_j) \tag{6}$$

In our experiment, $C = 0.986$ across all 4 centroids on the full 722K dataset ($C = 0.909$ on the 10K subset). This means the three malicious centroids are nearly indistinguishable from each other and from the benign centroid in cosine space. When centroids overlap this severely,

the argmax decision boundaries become essentially random among the 4 categories, yielding 40.2% accuracy on 722K and 68.3% on 10K—both insufficient for practical deployment.

When the three malicious categories are merged into a single class, the classification becomes:

$$\hat{y}_{\text{binary}} = \mathbb{1}\left[\cos(\mathbf{x}, \mu_{\text{malicious}}) > \cos(\mathbf{x}, \mu_{\text{benign}})\right] \tag{7}$$

The merged malicious centroid $\mu_{\text{malicious}} = \frac{1}{|S_{\text{mal}}|} \sum_{i \in S_{\text{mal}}} \mathbf{x}_i$ aggregates the shared representation of all three attack types while reducing category-specific variation. This improves accuracy to 71.8% (AUC 0.772), a 31.6 pp gain, because the binary decision boundary only needs to separate two clusters rather than resolve four overlapping ones. The centroid similarity C remains 0.986 (measured between the two remaining centroids), confirming that even with 2 centroids, the benign and malicious classes overlap substantially in cosine space—but the binary classification is robust enough to extract a usable boundary.

5.6 Interpreting the Centroid Paradox

The central finding of our investigation is a paradox: nearest-centroid scoring fails (AUC 0.547) while linear probes succeed (AUC 0.884) on the same embeddings. This paradox reveals a fundamental limitation of centroid-based classification in high-dimensional spaces.

A nearest-centroid classifier assigns $\hat{y} = \arg\max_c \cos(\mathbf{x}, \mu_c)$. This is equivalent to a nearest-neighbor classifier with $k = 1$ where the prototype for each class is its mean. The decision boundary between two classes i and j is the hyperplane:

$$\mathbf{w} = \mu_i - \mu_j, \quad \text{classify as } i \text{ if } \mathbf{x} \cdot \mathbf{w} > \frac{\|\mu_i\|^2 - \|\mu_j\|^2}{2} \tag{8}$$

This boundary depends only on the class means. When $C = \mu_i \cdot \mu_j \approx 1$ (as in our experiment where $C = 0.986$), the means are nearly collinear, and the boundary becomes extremely sensitive to small perturbations in the data. The classifier cannot exploit structure beyond the first moment of each class distribution.

A logistic regression linear probe learns a weight vector $\mathbf{w}^*$ that maximizes the likelihood of the training data:

$$\mathbf{w}^* = \arg\max_{\mathbf{w}} \sum_{i=1}^{n} \left[y_i \log \sigma(\mathbf{w}^\top \mathbf{x}_i) + (1 - y_i) \log(1 - \sigma(\mathbf{w}^\top \mathbf{x}_i)) \right] \tag{9}$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the sigmoid function. Unlike the centroid classifier, logistic regression optimizes $\mathbf{w}$ over all training examples simultaneously, finding a direction in the 1024-dimensional space that best separates the classes. It is not constrained to use the class means as prototypes.

In high-dimensional spaces ($d = 1024$), two classes can have nearly identical means ($C = 0.986$) while remaining linearly separable along other dimensions. The centroid classifier can only see the projection onto the $\mu_i - \mu_j$ direction, which is uninformative when $C \approx 1$. The linear probe can find a separating hyperplane in the full 1024-dimensional space, exploiting structure that is invisible to the centroid-based approach. This explains why linear probes achieve AUC 0.844–0.884 while nearest-centroid achieves only 0.547 on the same embeddings.

5.7 Why StandardScaler Helps

The Euclidean distance with StandardScaler outperforms cosine similarity (AUC 0.784 vs. 0.772) despite using the same 2-centroid classification. We analyze why through the lens of distance weighting.

StandardScaler transforms each dimension j to zero mean and unit variance:

$$x_j^{\text{scaled}} = \frac{x_j - \mu_j}{\sigma_j} \tag{10}$$

where μ_j and σ_j are the mean and standard deviation of dimension j across the training set. The squared Euclidean distance between two scaled points $\mathbf{x}$ and $\mathbf{y}$ is:

$$d^2(\mathbf{x}^{\text{scaled}}, \mathbf{y}^{\text{scaled}}) = \sum_{j=1}^{d} \left(\frac{x_j - y_j}{\sigma_j} \right)^2 = \sum_{j=1}^{d} \frac{1}{\sigma_j^2} (x_j - y_j)^2 \tag{11}$$

This is equivalent to a weighted Euclidean distance where each dimension j receives weight $w_j = 1/\sigma_j^2$. Dimensions with low variance (σ_j small) receive high weight, while high-variance dimensions are downweighted. In the context of bge-large-en-v1.5 embeddings, which are normalized to unit length, most dimensions have very similar variance across the dataset. However, dimensions that are discriminative between classes tend to have lower within-class variance—they are "stable" for each class but differ between classes. StandardScaler amplifies these dimensions by assigning them higher weight.

Cosine similarity, by contrast, normalizes only the vector magnitude:

$$\cos(\mathbf{x}, \mu_c) = \frac{\mathbf{x} \cdot \mu_c}{\|\mathbf{x}\| \|\mu_c\|} \tag{12}$$

This treats all dimensions equally (uniform weight $w_j = 1$), regardless of their discriminative power. StandardScaler's inverse-variance weighting provides a form of automatic feature selection, upweighting dimensions that vary little within classes but differ between them. This explains the observed AUC improvement from 0.772 (cosine) to 0.784 (Euclidean + StandardScaler) with the same 2-centroid classification method.

6 Hybrid Guardrail V3: Architecture

The hybrid guardrail combines three signal layers with a stacking ensemble classifier, designed to capture security-relevant patterns that general-purpose embeddings miss.

6.1 Feature Engineering (117 Features)

The model extracts 117 handcrafted features from each prompt, organized into 14 categories: basic statistics (6), character ratios (6), entropy (4), attack keywords (3), encoding detection (6), structural patterns (6), lexical repetition (3), structural flags (10), advanced structural (12), linguistic (12), readability (10), n-gram diversity (2), injection-specific (8), and punctuation (10).

6.2 Pattern Matching (Aho-Corasick)

An Aho-Corasick automaton (Aho and Corasick, 1975) matches 47 regex patterns in a single pass: $\text{score}_{\text{regex}}(t) = \max_{p \in \mathcal{P}} s_p$ if p matches t, else 0. Patterns include instruction overrides (0.95), role hijacking (0.90), system extraction (0.95), and delimiter injection (0.90).

6.3 TF-IDF Similarity (Quantized)

TF-IDF similarity (Manning and Schütze, 1999) was quantized to int8 for memory-efficient inference.

6.4 Stacking Ensemble

The stacking classifier combines XGBoost (500 trees, depth 6, lr 0.05), LightGBM (500 trees, depth 6, lr 0.05), and Random Forest (300 trees, depth 12) as base learners, with a logistic regression meta-learner, followed by isotonic calibration.

6.5 Ensemble Decision

The final ensemble combines the three signal layers:

$$\text{ensemble} = w_0 \cdot \text{score}_{\text{regex}} + w_1 \cdot \text{score}_{\text{sim}} + w_2 \cdot P_{\text{cal}}(x) \tag{13}$$

where $P_{\text{cal}}(x)$ is the calibrated stacking probability. The weights $[w_0, w_1, w_2]$ are initialized to $[0.10, 0.10, 0.80]$ and adapted online during inference via softmax:

$$w_i^{(t)} = \frac{\exp(a_i^{(t)})}{\sum_j \exp(a_j^{(t)})}, \quad a_i^{(t)} = \alpha \cdot a_{\text{comp},i}^{(t)} + (1 - \alpha) \cdot a_i^{(t-1)} \tag{14}$$

where $\alpha = 0.05$ and $a_{\text{comp},i} = [1 - \text{fp_rate}_i, 1 - \text{fn_rate}_i, \text{accuracy}_i]$ for each component. The stacking probability dominates because it achieves the highest individual accuracy; the adaptive mechanism reinforces this assignment. The regex and similarity signals remain at low weights because their individual discriminative power is lower.

6.6 Training Setup

- **Dataset:** 722,842 samples, 80/10/10 stratified split (578,273 train / 72,284 val / 72,285 test)
- **TF-IDF:** Word-level (5,000 features, bigrams, sublinear TF)
- **Stacking:** 3-fold CV, passthrough=True
- **Calibration:** Isotonic regression on validation set
- **Hardware:** Kaggle P100-PCIE-16GB, $\sim$5 hours training

7 Hybrid Guardrail V3: Results

7.1 In-Distribution Performance

Table 7: Hybrid Guardrail V3 test set performance.

Metric	Value
Accuracy	0.830
Precision (Weighted)	0.835
Recall (Weighted)	0.830
F1-Score (Weighted)	0.827
Confusion Matrix	
True Positives	38,543
True Negatives	21,734
False Positives	7,244
False Negatives	4,764

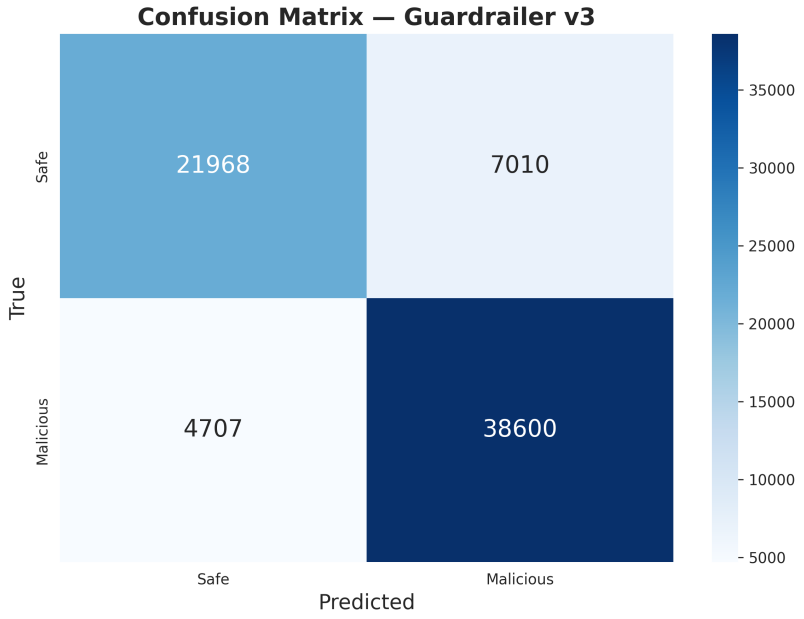


Figure 4: Confusion matrix for Hybrid Guardrail V3 on the test set. The model achieves 21,734 true negatives and 38,543 true positives, with 7,244 false positives and 4,764 false negatives.

The model achieves moderate recall (83.0%) with balanced precision. The AUC-ROC of 0.908 indicates strong ranking performance across thresholds on the in-distribution test set; however, at the selected operating threshold, the false-positive rate remains 25.0% of safe prompts misclassified, which would require reduction before deployment in user-facing settings.

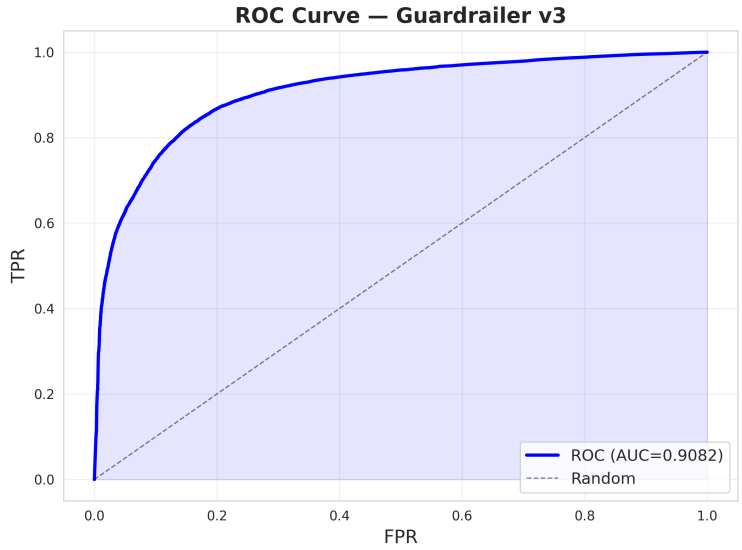


Figure 5: ROC curve for Hybrid Guardrail V3. The model achieves AUC = 0.908, indicating strong ranking performance across classification thresholds on the evaluated in-distribution test set.

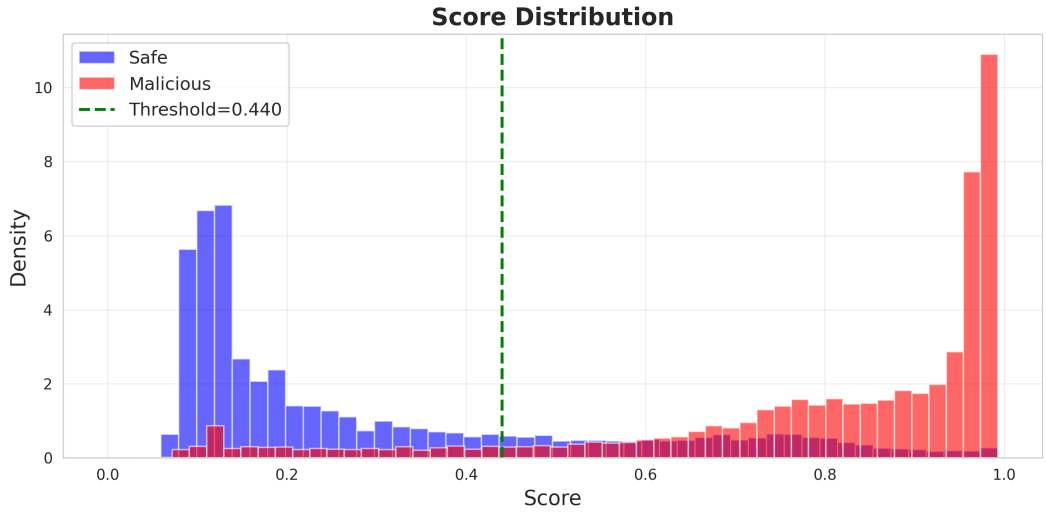


Figure 6: Score distribution for Hybrid Guardrail V3. The green dashed line at threshold = 0.460 separates safe (blue) and malicious (red) predictions. The distributions show substantial separation on the evaluated in-distribution test set, with overlap concentrated in the middle score range.

7.2 Transformation-Based Robustness Evaluation

Table 8: Transformation-based robustness evaluation: detection rate across 8 attack transformations ($n = 200$ malicious prompts).

Transform	Detection Rate	Status	Time
Original	92.5%	PASS	1.2s
Paraphrase	91.5%	PASS	1.3s
Misspelling	**63.5%**	**WARN**	1.3s
Unicode	83.5%	PASS	1.3s
Base64	100.0%	PASS	1.3s
Roleplay	96.5%	PASS	1.3s
Euphemistic	92.5%	PASS	1.3s
Long Context	93.5%	PASS	1.5s
Average	**88.7%**		

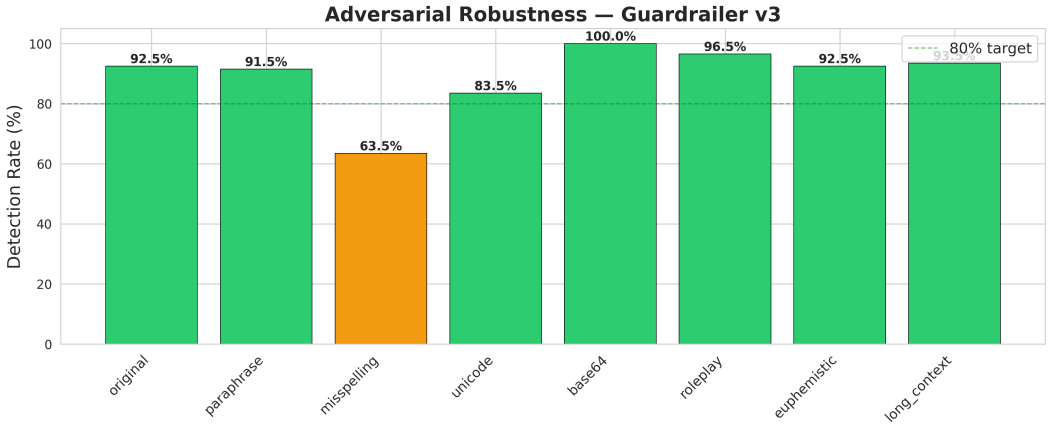


Figure 7: Transformation-based robustness evaluation across 8 attack transformations ($n = 200$). The green dashed line at 80% indicates the minimum acceptable detection rate. Misspelling (63.5%) is the only transformation falling below this threshold, while Base64 encoding achieves 100% detection. These results reflect fixed transformations only; adaptive adversarial attacks were not evaluated.

The model maintains detection rates $\geq 80\%$ across 7 of the 8 evaluated fixed transformations. Misspelling is the critical weakness (63.5%), where character substitution destroys word-level TF-IDF features. With $n = 200$ per transformation, these detection rates have approximate 95% confidence intervals of ± 5–7 percentage points; differences smaller than this margin should be interpreted cautiously. These results reflect robustness to fixed transformations only; the model was not evaluated against adaptive adversarial attacks that iteratively optimize to evade detection.

7.3 Cross-Dataset Generalization

Table 9: Cross-dataset generalization: model trained on Guardrailer Dataset v1, tested on four external datasets.

Dataset	Accuracy	F1	Samples	Δ Accuracy
In-distribution	**0.830**	**0.827**	72,285	—
JailbreakBench	0.550	0.670	4,350	−28.0 pp
Jailbreak Classification	0.590	0.680	4,350	−24.0 pp
Jailbreak Complete DS	0.422	0.571	4,350	−40.8 pp
JailbreakHub	0.496	0.220	4,350	−33.4 pp
Cross-dataset avg	**0.515**	**0.535**		**−31.5 pp**

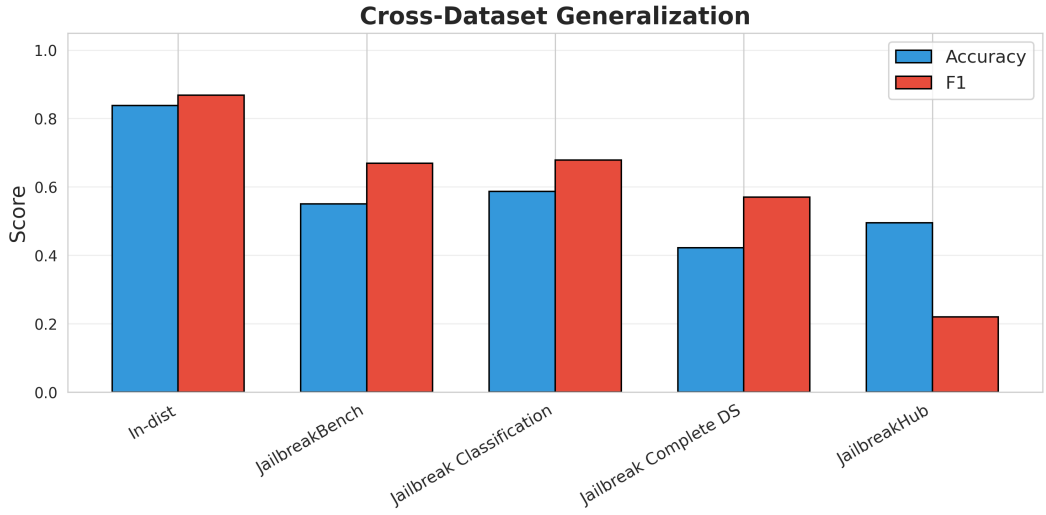


Figure 8: Cross-dataset generalization results. The model trained on Guardrailer Dataset v1 shows severe performance degradation on four external datasets. In-distribution accuracy (83.0%) drops to 42.2–59.0% on external benchmarks, with JailbreakHub showing particularly poor performance (F1 = 0.220).

This is the most important limitation identified by the external evaluation. The model achieves 83.0% in-distribution accuracy but drops to 51.5% on the four tested external distributions—a 31.5 percentage-point decline. Performance on Jailbreak Complete DS (42.2%) is below the 0.50 reference level for balanced binary classification, suggesting substantial distribution-specific failure on this benchmark. These results reflect performance under the tested distribution shifts; the external benchmarks do not represent all possible real-world prompt injections.

7.4 Feature Importance

Lexical repetition and diversity features were among the most important handcrafted features, capturing the template-like patterns common in jailbreak attempts.

7.5 Computational Requirements

Inference requires no GPU or vector database and achieved 0.149 ms/prompt (6,727 prompts/s) on a single-threaded CPU at batch size 1. Timing included TF-IDF vectorization, feature extraction, Aho-Corasick matching, ensemble prediction, and calibration, averaged over 10,000 runs after 1,000 warm-up iterations.

7.6 Ablation Study

To understand the contribution of each component, we conducted a focused ablation study on the Hybrid Guardrail V3. The study evaluates three core configurations—full model, TF-IDF only, and handcrafted features only—plus leave-one-feature-group-out variants. All experiments use the same dataset, 80/10/10 stratified split (578,273 train / 72,284 val / 72,285 test), stacking ensemble architecture, random seed (42), and calibration protocol.

Table 10: Ablation study results on 72,285 test samples. All models use the same stacking ensemble (XGBoost + LightGBM + RandomForest → LogisticRegression) with isotonic calibration.

Experiment	**Feat.**	**Acc.**	**Prec.**	**Rec.**	**F1**	**ECE**	**Thresh.**
Full baseline (TF-IDF + 117 HC)	5,117	0.830	0.835	0.830	0.827	0.0161	0.46
TF-IDF only	5,000	**0.850**	**0.853**	**0.850**	**0.848**	**0.0069**	0.46
Handcrafted only	117	0.800	0.802	0.800	0.797	0.0122	0.42
Full − text_statistics	5,111	0.850	0.847	0.850	0.847	0.0107	0.44

Feature counts refer to the input feature vector (117 handcrafted + 5,000 TF-IDF) before the ensemble's derived prediction features and calibration stage.

7.6.1 Key Findings

TF-IDF-only achieves 85.0% accuracy and F1=0.848, exceeding the full model (83.0%, F1=0.827) by 2.0 pp. Training and validation AUC were also higher for TF-IDF-only than the full configuration, consistent with the test-set ablation. Under the evaluated setting, the 117 handcrafted features did not provide complementary predictive value beyond TF-IDF. This result is specific to this dataset, model configuration, and experimental protocol; it does not demonstrate that handcrafted features are universally uninformative. Formal statistical significance testing was not conducted; the consistent direction of improvement across accuracy, F1, and ECE provides converging evidence for the finding.

8 Discussion

8.1 Main Findings and Practical Strengths

The central finding is that nearest-centroid scoring on general-purpose embeddings fails (AUC 0.547), yet the same embedding space contains discriminative information accessible to trained classifiers (AUC 0.844–0.884). This motivated a lightweight hybrid guardrail achieving 83.0% in-distribution accuracy at 0.149 ms/prompt (6,727 prompts/s) on CPU, with well-calibrated probabilities (ECE = 0.0161) and 100% Base64 detection.

8.2 Limitations and Failure Modes

Cross-dataset generalization failure. In-distribution accuracy drops from 83.0% to 51.5% (−31.5 pp) on four external benchmarks. Jailbreak Complete DS achieves 42.2% (below 0.50), and JailbreakHub achieves F1 = 0.220, indicating dataset-specific pattern memorization.

Misspelling vulnerability. Detection drops to 63.5% against character substitution, confirming reliance on character-exact TF-IDF matching.

High false-positive rate. At 25.0%, the FPR would generate unacceptable noise in user-facing settings.

Feature limitations. Under the evaluated setting, the 117 handcrafted features did not complement TF-IDF (ablation: TF-IDF-only 85.0% vs. full 83.0%). The recommended deployment is TF-IDF-only (ECE 0.0069 vs. 0.0161).

Limited robustness evaluation. The 8 tested transformations ($n = 200$) did not include adaptive adversarial attacks.

Embedding compute constraints. The linear probe, 2-centroid cosine similarity, and Euclidean + StandardScaler experiments (Table 6, Rows 2–5) were evaluated on 10,000 stratified samples due to the computational cost of generating and storing 1024-dimensional embeddings for the full 722K dataset. This highlights a practical limitation of embedding-based approaches: they require substantial compute for both training and inference compared to lightweight feature-based methods.

8.3 Practical Implications

Production deployment requires: (1) multi-source training data for generalization; (2) character-level normalization for misspelling robustness; (3) cost-sensitive training to reduce FPR; and (4) continuous monitoring for distribution drift.

9 Future Work

Key directions include: (1) multi-source training combining Guardrailer Dataset v1 with JailbreakBench, JailbreakHub, and other datasets; (2) character-level normalization via leetspeak-aware tokenization and Unicode NFKD decomposition; (3) semantic features using sentence transformer embeddings as additional inputs; (4) cost-sensitive thresholding to reduce the 25.0% false-positive rate; and (5) domain adaptation using gradient reversal to learn domain-invariant features.

10 Conclusion

This paper documents a systematic journey in prompt injection detection. The embedding method revealed that nearest-centroid scoring fails to achieve practical performance (AUC 0.547), suggesting that the evaluated general-purpose embedding models did not reliably expose security-relevant distinctions through the tested nearest-centroid scoring approach. Systematic investigation across embedding models shows the failure is primarily methodological: linear probes on the same embeddings achieve AUC 0.844–0.855, and reducing from 4 to 2 centroids substantially improves performance. We evaluated a lightweight hybrid guardrail (117 handcrafted features + TF-IDF + stacking ensemble) achieving 83.0% in-distribution accuracy at 0.149 ms/prompt. An ablation study revealed TF-IDF alone

achieves 85.0%, exceeding the full hybrid by 2.0 pp, indicating handcrafted features do not provide complementary value under the evaluated setting. Cross-dataset evaluation reveals a severe generalization gap (31.5 pp decline to 51.5%). The most critical open challenge remains out-of-distribution generalization, requiring multi-source training data and evaluation protocols that test robustness under distribution shift.

Data and Code Availability

The complete source code, trained models, and experimental artifacts are publicly available at:

- **GitHub:** https://github.com/dev-prashanna/Prompt_Injection_Detection
- **Kaggle:** https://www.kaggle.com/datasets/prashannadeveloper/guardrailer-dataset-v1

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